Azure SQL and Managed Instance

A Beginner-to-Practitioner Guide, From Scratch

Who this is for: DBAs and developers who know SQL Server (or even just basic SQL) and want to understand the Azure SQL family and Azure SQL Managed Instance (SQL MI) from the ground up.

What you will learn: cloud basics, the Azure SQL options, how SQL MI is built, networking, tiers, security, high availability, backup, migration, monitoring, cost control, and a 4-week learning plan with hands-on labs.

Note: Azure changes quickly. Limits, SKUs and prices in this guide are indicative. Always confirm on Microsoft Learn and the Azure pricing calculator before making decisions.

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1. Cloud and Azure basics

Before Azure SQL, understand four ideas that appear everywhere in Azure.

Concept	Meaning	Why it matters for SQL
Subscription	The billing and access boundary for your resources.	Your SQL resources and costs live inside a subscription.
Resource group	A logical folder that holds related resources.	Keep a MI, its VNet and its monitoring together.
Region	A geographic Azure location (e.g. Central India, South India).	Choose near users and DR pair for failover.
VNet / subnet	Your private network inside Azure.	MI lives inside your VNet, so network design is essential.
Microsoft Entra ID	Azure's identity service (formerly Azure AD).	Modern, passwordless sign-in to SQL.

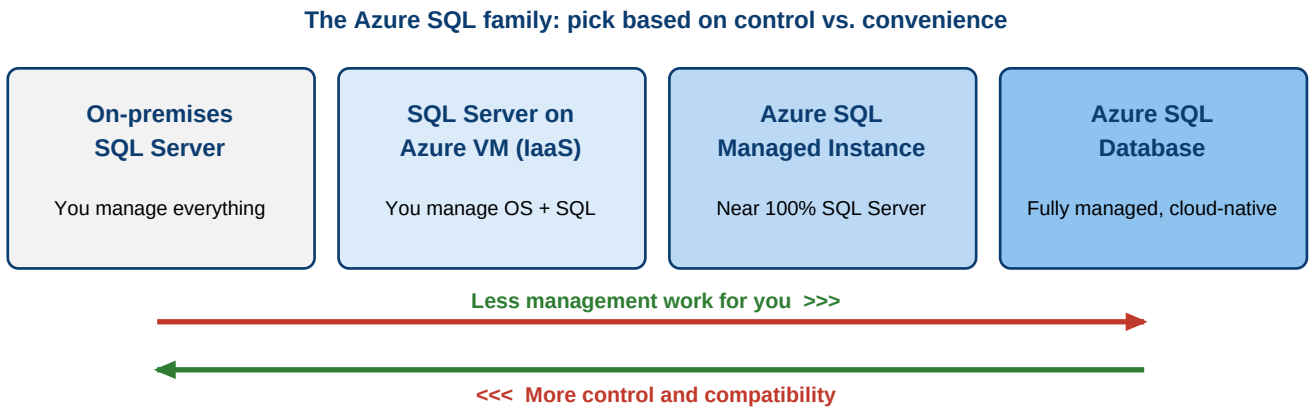
IaaS vs PaaS vs SaaS

IaaS (Infrastructure as a Service) gives you a virtual machine. You patch the OS, install SQL Server, configure backups and HA yourself. **PaaS** (Platform as a Service) gives you a running database service. Microsoft handles hardware, OS, SQL patching, backups and HA; you manage data, schema, queries and security design. Azure SQL Database and Managed Instance are **PaaS**.

Shared responsibility

Responsibility	On-prem	SQL on Azure VM	Managed Instance	SQL Database
Hardware and datacenter	You	Microsoft	Microsoft	Microsoft
OS patching	You	You	Microsoft	Microsoft
SQL engine patching	You	You	Microsoft	Microsoft
Backups	You	You	Microsoft (automatic)	Microsoft (automatic)
High availability	You	You	Built-in	Built-in
Schema, queries, indexes	You	You	You	You
Security design and access	You	You	You (shared)	You (shared)

2. The Azure SQL family



Option	What it is	Best for
Azure SQL Database	Cloud-native database. Deployed as a single database or in an elastic pool. Includes serverless and Hyperscale options.	New cloud apps, SaaS multi-tenant, microservices, variable workloads.
Azure SQL Managed Instance	A managed SQL Server instance in your VNet with near 100% compatibility with the on-premises SQL Server engine.	Lift-and-shift of existing SQL Server apps with minimal code changes.
SQL Server on Azure VM	Full SQL Server on a virtual machine. Total control.	Apps needing OS access, SSRS/SSAS/SSIS on the same box, unsupported features, specific versions.

Quick decision guide

- Building a **new** cloud-native application? Start with **Azure SQL Database**.
- Moving an **existing** SQL Server app that uses SQL Agent, cross-database queries, linked servers, CLR or Service Broker? Choose **Managed Instance**.
- Need OS-level access, a specific SQL version, or third-party agents on the server? Choose **SQL on Azure VM**.

3. Azure SQL Database in brief

You need a basic view of Azure SQL Database because it is the sibling of MI and often compared with it.

Deployment	Description
Single database	One isolated database with its own resources and its own logical server for login and firewall settings.
Elastic pool	Many databases share a pool of resources. Good for many small databases with unpredictable, non-overlapping peaks.
Serverless compute	Auto-scales and can auto-pause when idle. You pay for compute used per second.
Hyperscale	Separates compute and storage. Supports very large databases (up to about 128 TB) with fast scaling and quick backups.

Purchasing models: **vCore** (choose CPU, memory and storage separately; recommended) and **DTU** (a bundled unit, simpler but less flexible).

4. Azure SQL Managed Instance (the core)

Azure SQL Managed Instance is a fully managed PaaS service that runs the latest SQL Server database engine, with **instance-scoped features**. Think of it as 'SQL Server as a service' inside your own private network.

What you get

- An **instance**, not just a database: multiple user databases plus system databases (master, msdb, model, tempdb).
- **SQL Server Agent** for scheduled jobs and alerts.
- **Cross-database queries and transactions** within the instance.
- **Linked servers**, **CLR**, **Service Broker**, **Database Mail**, and **native backup/restore from a .bak file** using Azure Blob Storage.

- Private IP inside your VNet, so it can be reached from on-premises over VPN or ExpressRoute.
- Automatic patching, automatic backups, built-in high availability, and Microsoft Entra authentication.

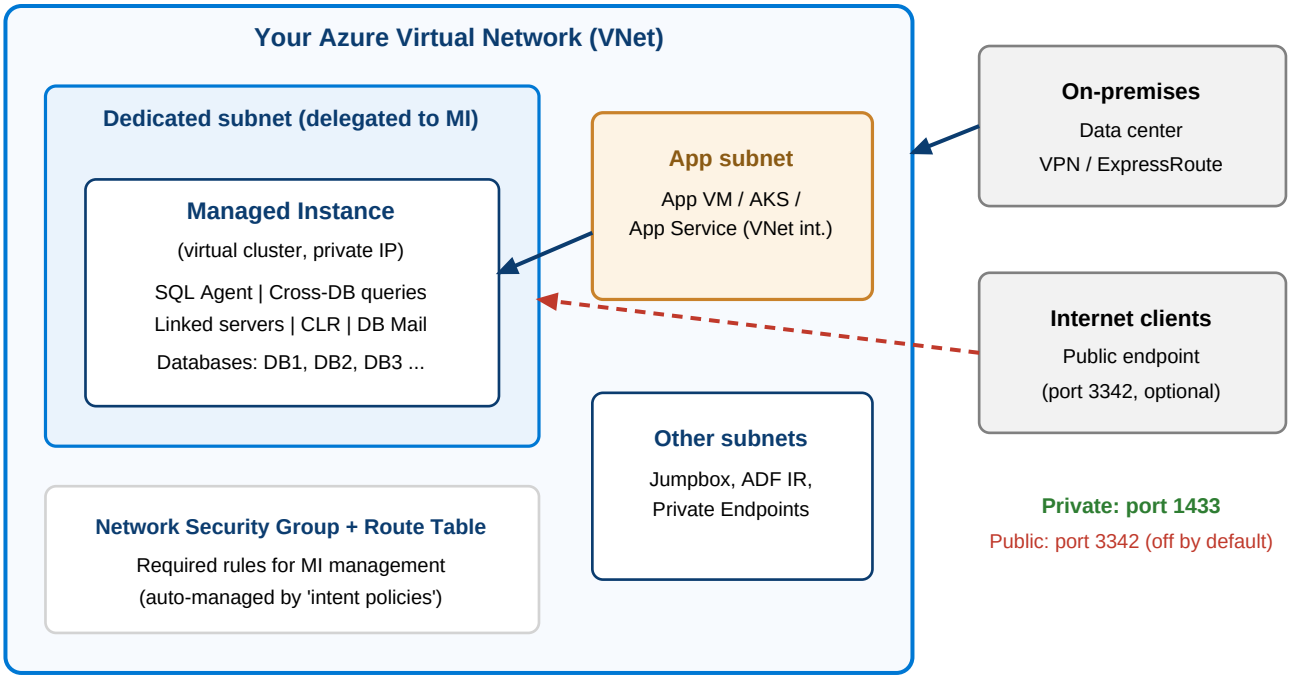
What is different from on-prem SQL Server

- No access to the OS or file system. Files are managed by Azure; you do not choose file paths.
- No FILESTREAM and some features that depend on the OS or hardware.
- **Trace flags, sp_configure, and some server settings** are limited or controlled by the service.
- Backups are automatic; you cannot disable them. You can add copy-only backups to URL.
- Patching and maintenance happen on Microsoft's schedule, with a configurable maintenance window.

Tip: Microsoft's feature comparison page lists T-SQL differences between SQL Server and SQL MI. Run the Azure Migrate or Data Migration Assistant assessment before moving, so you see blockers early.

5. MI architecture and networking

Unlike Azure SQL Database, MI is **injected into your virtual network**. It sits in a dedicated, delegated subnet and gets a private IP. Behind the scenes it runs on a **virtual cluster** of compute nodes that Microsoft manages.



Networking essentials

Topic	What to know
Dedicated subnet	The subnet can host only managed instances. Size it with room to grow (a /27 is the minimum; plan larger if you will scale or add instances).
Subnet delegation	The subnet is delegated to Microsoft.Sql/managedInstances. Azure then manages the required network rules for you.
Private endpoint (VNet-local)	Default access: private IP, port 1433, with a host name like `myinstance.xxxx.database.windows.net`.

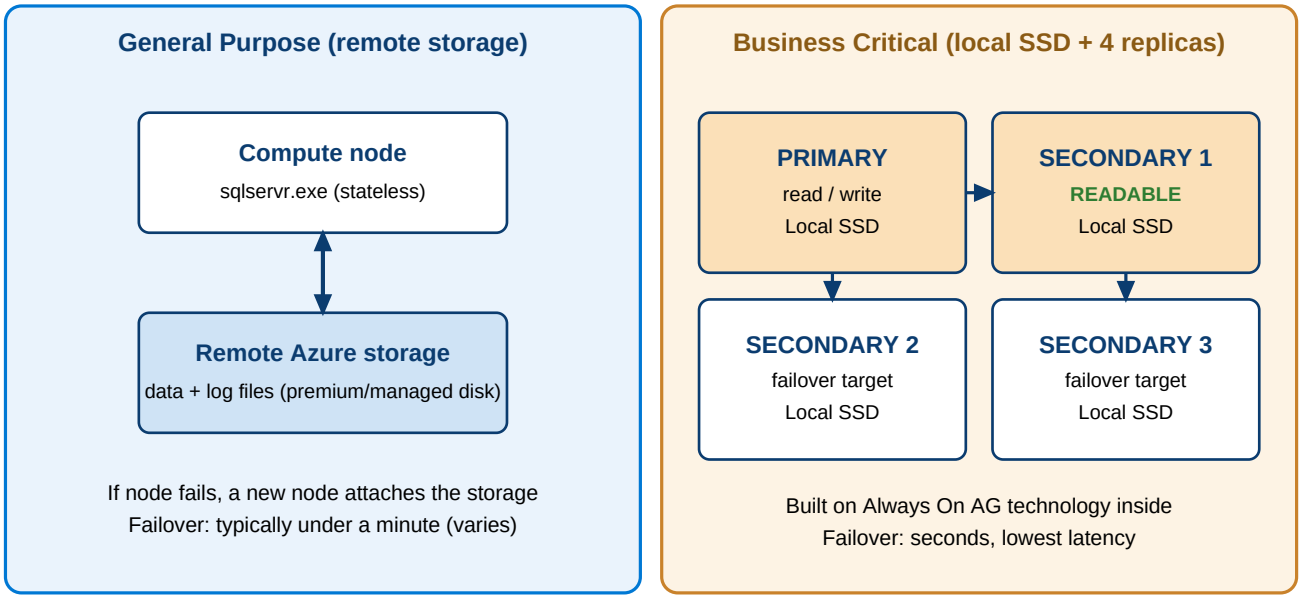
Topic	What to know
Public endpoint	Optional. Uses port 3342. Off by default. If enabled, restrict it with NSG rules to known IP ranges.
Connectivity to on-prem	Site-to-site VPN or ExpressRoute into the VNet. Plan DNS resolution for the MI host name.
Connection types	Proxy or Redirect. Redirect is usually faster but needs more ports opened inside the VNet.

Compute behind the scenes

A **virtual cluster** is a set of dedicated VMs for your subnet. The first instance in a new subnet takes the longest to create because the cluster is built; later instances in the same subnet are faster. Resizing compute can also involve moving the instance between nodes, so expect a short failover at the end.

6. Service tiers and compute options

The tier defines your storage architecture, performance, and resilience.



Feature	General Purpose (classic)	Next-gen General Purpose	Business Critical
Storage	Remote storage (page blobs)	Remote storage on managed disks, higher IOPS and throughput	Local SSD on each node
Max storage	Up to 16 TB	Up to 32 TB	Up to 16 TB (hardware-dependent)
Max databases	100	Higher limit (up to 500)	100
Storage latency	Higher (about 5 to 10 ms)	Improved	Lowest (sub-millisecond to low ms)
Built-in replicas	1 (stateless compute, storage is remote)	1 (stateless compute, remote storage)	4 nodes: 1 primary + 3 secondaries
Readable secondary	No	No	Yes (1 included, free)

Feature	General Purpose (classic)	Next-gen General Purpose	Business Critical
Failover	Typically under a minute, varies	Typically under a minute, varies	Fastest (seconds)
Relative cost	Lowest	Same billing as General Purpose	Roughly 2.7x General Purpose
Best for	General business workloads	Most workloads wanting more performance without BC price	Latency-sensitive OLTP, in-memory OLTP, heavy read offload

Next-gen General Purpose reached general availability in November 2025 as an upgrade to the existing General Purpose tier. It is billed as General Purpose but offers improved storage performance, flexible memory options, and larger capacity. For new projects, evaluate Next-gen GP first, and move to Business Critical only if you truly need very low storage latency or a free readable secondary.

Other choices

- **vCore model:** choose vCores (and memory in newer options) and storage independently. Hardware series include Standard-series (Gen5) and Premium-series options for more memory per vCore.
- **Update policy:** 'Always-up-to-date' gets the newest engine features sooner, while a fixed SQL Server version policy (for example SQL Server 2022 policy) keeps engine features aligned with that version. Choose based on your compatibility needs.
- **Instance pools:** run several small instances (down to 2 vCores) on shared resources to lower cost for many small workloads.
- **Azure Hybrid Benefit:** use existing SQL Server licenses with Software Assurance to reduce compute cost.
- **Reserved capacity:** commit for 1 or 3 years for a lower rate.

7. Creating your first Managed Instance

Before you start (checklist)

1. An Azure subscription with permission to create network and SQL resources.
2. A VNet with a dedicated, empty subnet (/27 or larger) delegated to Microsoft.Sql/managedInstances.
3. Region, tier (GP / Next-gen GP / BC), vCores, storage size, and collation decisions.
4. An identity plan: Microsoft Entra admin (preferred) and an optional SQL admin.
5. Time. The first creation in a subnet can take several hours.

Azure portal path

1. Search for **Azure SQL Managed Instance** and choose Create.
2. Select subscription, resource group, name, region.
3. Choose compute + storage (service tier, hardware, vCores, storage).
4. Choose networking (VNet and subnet; public endpoint off for production).
5. Set authentication and security options, then Review + create.

Azure CLI example

```
# Variables
RG=rg-sqlmi-demo
LOC=centralindia
VNET=vnet-sqlmi
SUBNET=snet-sqlmi
MI=sqlmi-demo-001
SUBNET_ID=$(az network vnet subnet show -g $RG \
  --vnet-name $VNET -n $SUBNET --query id -o tsv)

az group create -n $RG -l $LOC

az sql mi create \
  --name $MI --resource-group $RG --location $LOC \
  --admin-user sqladmin --admin-password '<StrongPassword>' \
  --subnet $SUBNET_ID \
  --capacity 8 --edition GeneralPurpose --family Gen5 \
  --storage 256GB --license-type BasePrice
```

Connect and run a first query

- Use SSMS or Azure Data Studio from a VM in the same VNet (a small jumpbox VM works well for labs).
- Server name: `sqlmi-demo-001.<dnszone>.database.windows.net`.
- Sign in with Microsoft Entra or the SQL admin.

```
SELECT @@VERSION;
SELECT name, compatibility_level, state_desc FROM sys.databases;
CREATE DATABASE SalesDB;
```

8. Security

Security in Azure SQL follows a layered model: network, identity, data protection, and monitoring.

Layer	Capabilities
Network	Private IP in VNet, NSGs, Private Endpoint, public endpoint off by default, firewall via NSG rules.
Identity	Microsoft Entra authentication (MFA, Conditional Access, managed identities), Entra-only mode to disable SQL logins, server roles and RBAC.
Data at rest	Transparent Data Encryption (TDE) is on by default; optional customer-managed keys in Azure Key Vault.
Data in transit	TLS encryption enforced; set a minimum TLS version.
Data in use	Always Encrypted, row-level security, dynamic data masking, column-level permissions.
Threat protection	Microsoft Defender for SQL (vulnerability assessment and threat detection).
Auditing	SQL Audit to Storage, Log Analytics, or Event Hubs.

Create an Entra user in the database

```
-- Connect as a Microsoft Entra admin, then:
CREATE USER [dba.team@contoso.com] FROM EXTERNAL PROVIDER;
ALTER ROLE db_datareader ADD MEMBER [dba.team@contoso.com];

-- For an app using a managed identity:
CREATE USER [my-app-identity] FROM EXTERNAL PROVIDER;
```

Best practice: prefer Entra authentication and managed identities over SQL passwords, grant least privilege, and keep the public endpoint off unless you have a clear need.

9. High availability and disaster recovery

High availability (HA) keeps you running during local failures. **Disaster recovery (DR)** keeps you running if a whole region fails.

Level	Mechanism	Notes
Local HA (GP)	Stateless compute with remote storage. A failed node is replaced and reattached to storage.	Brief interruption during failover. Retry logic in your app is vital.
Local HA (BC)	Four-node Always On AG with local SSD.	Fast failover; one secondary is readable.
Zone redundancy	Spread replicas across availability zones in a region.	Protects against datacenter (zone) failure. Check supported regions.
Geo-DR	Auto-failover groups to a secondary instance in another region.	Read-write and read-only listener endpoints that do not change after failover.

Auto-failover group (geo-replication) at a glance

- Create a secondary MI in another region (often a paired region) with the **same service tier and compatible sizing**.
- Create a failover group. Azure replicates databases asynchronously.
- Applications connect through the group's **read-write listener**. After a failover the listener points to the new primary, so connection strings do not change.
- The secondary can serve read-only workloads through the read-only listener.
- Because replication is asynchronous, RPO is typically seconds; test failovers regularly.
- Both instances normally need connectivity (VNet peering or gateways) and DNS configured for the group.

Compare with the DAG you learned earlier: on SQL Server you build a Distributed Availability Group manually. On Azure SQL MI, the **auto-failover group** gives a similar cross-region async design as a managed feature, with listener endpoints handled for you.

SLA and retry logic

Azure SQL MI is backed by an availability SLA (check the current SLA document). Regardless of tier, brief connection drops can occur during maintenance and failover. Always implement **connection retry with exponential backoff** in your application.

10. Backup and restore

Backups are **automatic**: full backups weekly, differential backups every 12 to 24 hours, and transaction log backups about every 10 minutes. These are stored in redundant Azure storage and enable point-in-time restore (PITR).

Capability	Details
Point-in-time restore (PITR)	Restore a database to any second within the retention window. Default retention is 7 days and can be set from 1 to 35 days.
Long-term retention (LTR)	Keep weekly, monthly or yearly full backups for up to 10 years for compliance.

Capability	Details
Backup storage redundancy	Locally redundant, zone-redundant or geo-redundant. Geo-redundant enables geo-restore to another region.
Deleted database restore	You can restore a dropped database within the retention period.
Copy-only backup to URL	Manual copy-only backups to Azure Blob for offline copies or migration back to SQL Server.
Native restore from URL	Restore on-prem .bak files from Blob Storage into MI.

Restore examples

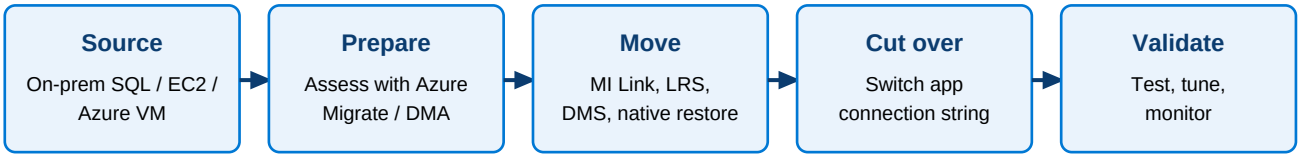
```
-- Native restore of an on-prem backup stored in Azure Blob (using managed identity or SAS)
RESTORE DATABASE SalesDB
FROM URL = 'https://mystorage.blob.core.windows.net/backups/SalesDB.bak';

-- Copy-only backup to URL
BACKUP DATABASE SalesDB
TO URL = 'https://mystorage.blob.core.windows.net/backups/SalesDB_copy.bak'
WITH COPY_ONLY, COMPRESSION;
```

Point-in-time restore is done from the Azure portal, CLI or PowerShell. It creates a **new database** on the same instance (or another instance), and does not overwrite the original.

11. Migrating to Managed Instance

Typical migration flow to Azure SQL Managed Instance



Step 1: Assess

Use **Azure Migrate** or the **Azure Data Migration extension for Azure Data Studio** to find compatibility issues and get a recommended SKU based on performance data.

Step 2: Choose a migration method

Method	How it works	Downtime	Best for
Managed Instance link	Near real-time replication from SQL Server to MI using distributed availability group technology. Cut over when ready.	Very low (minutes)	Large or busy databases, minimal downtime, ability to test with read workloads.
Log Replay Service (LRS)	You take full, differential and log backups to Blob Storage; LRS continuously restores them into MI.	Low (minutes)	Offline-style replication where you control the backup chain.
Azure Database Migration Service (DMS)	Managed service that orchestrates backup/restore based migration, offline or online.	Offline: longer. Online: low.	Guided, portal-driven migrations of several databases.

Method	How it works	Downtime	Best for
Native backup/restore	Take a .bak, upload to Blob, RESTORE ... FROM URL.	Longer (depends on size)	Small databases, simple one-time moves.
BACPAC / export-import	Schema + data export, then import.	Longest	Small databases only; not recommended for large ones.

Step 3: Cut over and validate

- Stop writes to the source, apply the final backup or sync, and complete the cutover.
- Update application connection strings (or use a DNS alias to make this a one-line change).
- Recreate or verify **logins, SQL Agent jobs, linked servers, and Database Mail** because they live at the instance level.
- Run performance baselines and compare with the source. Update statistics and consider the database compatibility level.
- Keep the source available for a rollback window.

Remember the MI link: because it uses distributed availability group technology, your DAG knowledge from SQL Server applies directly. The MI link also works in reverse for supported scenarios, which makes it useful for hybrid DR and read-scale-out designs.

12. Monitoring and performance

Tool	Use
Azure Monitor metrics	CPU %, storage used, IO, memory usage, instance health. Create alerts on thresholds.
Log Analytics + diagnostic settings	Send SQL logs, errors, and audit data for central querying and alerts.
Query Store	On by default. Tracks query performance history and plan changes. Use it to find regressions and force good plans.
Intelligent Query Processing	Engine features (adaptive joins, memory grant feedback, PSP optimization, etc.) enabled by compatibility level.
Dynamic Management Views	Use the same DMVs you know (sys.dm_exec_query_stats, sys.dm_os_wait_stats, sys.dm_io_virtual_file_stats).
Defender for SQL	Security alerts and vulnerability assessment.

```
-- Top queries by CPU
SELECT TOP 10 qs.total_worker_time/qs.execution_count AS avg_cpu,
       qs.execution_count, SUBSTRING(st.text,1,200) AS query_text
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
ORDER BY avg_cpu DESC;

-- File IO latency (important on General Purpose)
SELECT DB_NAME(vfs.database_id) db, mf.physical_name,
       vfs.io_stall_read_ms/NULLIF(vfs.num_of_reads,0) AS avg_read_ms,
       vfs.io_stall_write_ms/NULLIF(vfs.num_of_writes,0) AS avg_write_ms
FROM sys.dm_io_virtual_file_stats(NULL,NULL) vfs
JOIN sys.master_files mf ON vfs.database_id=mf.database_id AND vfs.file_id=mf.file_id;
```

- **On General Purpose**, file size affects IO performance in classic GP, so pre-size files appropriately; Next-gen GP improves this.
- Enable **Query Store** capture and review top resource consumers weekly.
- Set **alerts** for high CPU, low storage, and failed failovers.

13. Cost optimization

Lever	How it helps
Right-size vCores and storage	Start from measured usage, not on-prem hardware size. Scale as needed.
Pick the right tier	Use Next-gen GP or GP unless BC is truly required. BC costs roughly 2.7x General Purpose.
Azure Hybrid Benefit	Apply existing SQL Server licenses with Software Assurance.
Reserved capacity	1 or 3 year commitments for steady workloads.
Instance pools	Share compute across several small instances.
Dev/test pricing	Use dev/test subscription pricing for non-production.
Free offer	Azure offers a free trial MI option for learning; check the current terms and limits.
Backup retention	Keep PITR retention only as long as needed; limit LTR to compliance needs.
Delete idle test instances	MI bills while it exists, so delete lab resources when done.

14. Common pitfalls

- **Subnet too small or shared:** plan the subnet before creation; mixing other resources is not allowed.
- **Forgetting DNS:** on-prem clients must resolve the MI host name to the private IP.
- **Assuming 100% feature parity:** run an assessment. Some features differ or are unavailable.
- **No retry logic:** brief failovers during maintenance can break apps without retries.
- **Logins and jobs not migrated:** they are instance-level objects and do not move with databases.
- **Public endpoint left open:** use private access and NSG rules.
- **Choosing BC by default:** test Next-gen GP first to save cost.
- **Long creation time surprise:** first instance in a subnet can take hours; schedule accordingly.
- **Ignoring collation:** instance collation is set at creation and cannot be changed later.

15. 4-week learning plan and cheat sheet

Week	Focus	Hands-on lab
1	Azure basics, Azure SQL family, Azure SQL Database	Create an Azure free account, deploy a single Azure SQL Database, connect with SSMS or Azure Data Studio, restore a sample database (e.g. AdventureWorks).
2	Managed Instance architecture, networking, tiers	Create a VNet and delegated subnet, deploy a small MI (or use the free offer), build a jumpbox VM, connect and create databases, SQL Agent job.
3	Security, backup, restore, monitoring	Add a Microsoft Entra admin and users, enable auditing to Log Analytics, run PITR, native restore from Blob, review Query Store.
4	HA/DR and migration	Create an auto-failover group or MI link in a lab, perform a test failover, run an assessment and a sample LRS/DMS migration.

Quick cheat sheet

Question	Answer
What is SQL MI?	Managed SQL Server instance in your VNet, near 100% compatible.
When to choose MI over SQL Database?	When you need instance-scoped features (Agent, cross-DB, linked servers, CLR, etc.) or easy lift-and-shift.
GP vs BC?	GP: remote storage, lower cost. BC: local SSD, 4 replicas, fastest, readable secondary.
What is Next-gen GP?	Upgraded General Purpose tier with better storage performance and capacity, billed as GP.
Where does MI live?	In a dedicated delegated subnet of your VNet.
Default ports?	1433 (private), 3342 (public endpoint).
How to do DR?	Auto-failover group to another region.
Best migration with low downtime?	MI link or Log Replay Service.
Backups?	Automatic, PITR 1 to 35 days (default 7), LTR up to 10 years.

Official resources to bookmark

- Microsoft Learn: Azure SQL Managed Instance documentation (learn.microsoft.com/azure/azure-sql/managed-instance)
- Azure pricing calculator and the Azure SQL MI resource limits page
- Azure Updates and the Azure SQL Blog for new features
- Microsoft Learn learning paths for Azure SQL and DP-300 (Administering Azure SQL Solutions)